



An exploratory analysis of the Small Modular Reactor ecosystem

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Abstract

This report provides an overview on the state of play for small modular reactors (SMRs) in a techno-economic perspective by presenting a preliminary mapping of the SMR ecosystem using the Techno-Economic ecoSystem (TES) approach. This methodology allows for a comprehensive mapping of the complex ecosystem constituting the SMR industrial landscape. Through the identification of activities and public and private actors interplaying, we can outline the EU strategic positioning as well as interdependencies, vulnerabilities, and opportunities for the development of these types of advanced reactors. The study offers insights on the EU role in the SMR network through collaboration and participation in the information flows.

The main findings exhibit the critical need to transform European research into innovation to stay at the competitive forefront. A well-developed research network is a valuable asset to keep up with SMR technological progress, but the EU needs to foster its business network. Observing SMR activities that are related to other industrial ecosystems, we highlight the importance of these reactors to support the development and decarbonisation of energy-intensive industries and meet other objectives of the Clean Industrial Deal.

While the present study offers a first overview of the results of the SMR techno-economic ecosystem mapping, the TES approach allows for exploring further analytical avenues and fine-tuning classification criteria. This novel analysis could provide strategic guidance to support public policies and international collaboration initiatives for further SMR development and deployment.

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Executive summary

The world needs more power. The International Energy Agency (IEA) estimates that global energy demand will increase more than 6% –or around 0.5% annually– by 2035 compared to 2023 levels (IEA, 2024b). At the same time, current geopolitical tensions and the political pledge to contain global warming have led to a push for a clean energy transition. Reducing fossil fuel dependency is not only environmentally necessary but also relevant for strategic autonomy. Nuclear energy can generate clean power, combine hybrid energy systems with renewable sources, provide baseload power to stabilise the electricity grid, and increase energy security through locally-produced SMR power that reduces exposure to global energy markets. These capacities are providing new nuclear technologies with momentum (IEA, 2022).

Small modular reactors, or SMRs, are advanced nuclear reactors that incorporate state-of-the-art features into production and safety systems. There is growing interest around the globe in SMR technology owing to its flexibility, adaptability and low cost. They can be mass produced and provide up to 300 megawatts of electric capacity per unit, equivalent to one third of traditional nuclear reactors.

According to the Nuclear Energy Agency (NEA), there are at least five operational SMRs –at least two known reactors in Russia, two in China and an experimental one in Japan–in the world (NEA, 2024b). Even in the most conservative scenario with the current policy settings, cumulative investment estimates in SMRs amount to USD 30 billion by 2050 (IEA, 2025b). This capital would translate in an increased deployed SMR capacity of 40 GW by 2050. In the EU, there are 15 countries with active interest in SMRs. A growing number of countries outside the EU also share an interest in the development of SMRs. The US, Canada, China, Japan, Russia and the UK are the most advanced countries in the development of SMRs, although most of their projects are in the design phase. Many developing nations such as India have also expressed interest in this technology.

Policy context

The Draghi report outlined three major actions as the EU's roadmap for sustainable growth: closing the innovation gap with China and the US, decarbonising the economy while keeping up with global competitiveness, and reducing dependencies (Draghi, 2024). To achieve these objectives, it is necessary to renovate the EU common energy market.

SMRs offer opportunities to contribute addressing these objectives. Europe needs energy to power up the digital transformation and close the innovation gap. These advanced reactors provide clean energy and offer the possibility to combine with other renewable energy sources, providing stable energy supply in the path towards a clean Europe. In addition, their scalability, modularity and size make them a suitable option to reduce energy costs and reach isolated areas where energy is expensive. For example, large nuclear plants impose sizeable construction costs. Investors must often bear the bulk of these sunk construction and performance investment risks. SMRs permit a more incremental investment than larger units of the past. They also claim to be a particularly suitable and affordable option for non-electrical applications such as hydrogen production (IAEA, 2024c).

This report presents the state of play in the SMR techno-economic ecosystem at the global and EU level, with the aim of supporting informed policy decisions on how to better support SMR development and deployment in the context of a new Clean Industrial Deal and productivity boost (European Commission and Leyen, 2024).

Key conclusions

- SMRs have the capacity to contribute to achieve EU's strategic economy objectives, as they offer a clean, flexible and affordable solution to propel energy generation.
- The SMR ecosystem comprises nearly 4,000 players, 85% of whom are private firms. The rest of the ecosystem includes about 500 research institutes and a relatively small number of government institutions.
- In the EU, the number of private firms comprises less than 80% of the total number of players. This share drops even further when excluding participants in EU-funded projects.
- The share of EU business activities (excluding EU-funded activities) is in line with the US and other Western countries, and larger than in Russia and Asian countries. The EU also dedicates more activities to research than the global average. However, the share of innovation activities in the EU is below 60%.
- At the global level, the focus is on innovation activities. Apart from the UK, more than 75% of the activities of all major EU competitors such as China and the US are dedicated to innovation. UK players allocate more activities into both business and research. The activity composition depicts an industry that prioritises the protection of innovation and the licencing of commercial rights for concepts and technological developments prior to production and commercialisation.
- Firms develop and register most patents on their own. Only one in four innovation activities is collaborative, and science-to-business collaboration accounts for less than 4% of patenting activities.
- High-temperature gas-cooled reactors (HTGR) and pressurised water reactors (PWR) account for the highest number of activities among SMR models. Pressure vessels are the most frequently captured SMR component.
- The EU stands out for the capacity of its research network. The EU leads in scientific publications, and collaborations within the SMR field. The EU also generates more autonomous research than any other region. Challenges lie in the EU's ability to transform its research excellence into innovation and to expand its business network.
- Member State participation in the European SMR landscape differs, as a large pool of players are in countries with a positive position on SMRs. EU-wide public support to SMRs would help optimise EU networks, resources, and knowledge to develop a competitive SMR ecosystem.
- The number of foreign-controlled EU firms has not increased since 2018 despite a growing number of businesses. Out of the 40 EU firms controlled by non-EU countries, American capital dominates (45%), whereas China controls three firms.
- The network analysis of the SMR ecosystem confirms a well-connected European scientific network. However, the EU needs to strengthen international scientific collaboration with leading institutions to play a strategic role in frontier research.
- The EU has limited influence in innovation knowledge flows across the SMR ecosystem. Similar to research, the EU needs to establish strategic partnerships in patenting activities as a previous step to commercial development.
- SMRs have a potential to play a key role in supporting energy-intensive industries.

- While SMRs can help to meet energy demand from the digital transformation, they can also benefit from innovation in areas such as AI and cybersecurity.

Related and future JRC work

The European Commission Joint Research Centre has a portfolio of activities dedicated to small modular reactors. Portfolios bring together scientific disciplines from across the JRC and, through a cross-disciplinary approach, create new insights, new knowledge and new methods to better coordinate policy support. The SMR portfolio helps EU Member States to assess the benefits of SMRs, examine the capacity and long-term sustainability of the supply chain for components and critical materials, and evaluate the impact on decarbonisation, the industry, and energy security. Future related work would further explore activities and players, considering a selection of categories, and examine firm characteristics.

1 Introduction

Nuclear policy has been present since the very beginning of the European Union. The six founding nations signed the Treaty on the European Atomic Energy Community, or Euratom Treaty, in 1957¹. It is one of the three founding treaties establishing the EU and all Member States are party to it. Over the years, the Euratom Treaty has remained largely unchanged. Euratom secondary legislation establishes common rules and standards on the use of nuclear energy – Article 194(2) of the Treaty on the Functioning of the EU²–. However, each Member State independently decides whether to include nuclear energy in its own energy mix.

Currently 12 out of 27 EU Member States – Belgium, Bulgaria, Czechia, Finland, France, Hungary, The Netherlands, Romania, Slovakia, Slovenia, Spain, and Sweden – host nuclear power plants in their territories. Furthermore, other countries such as Poland are proposing to develop nuclear energy for the first time. According to Eurostat data, nuclear energy made up 16.4% of the EU energy mix in 2024, accounting for 23.4% of the total electricity produced³.

The EU's ambition to reach net-zero emissions by 2050 is a major challenge for incoming and future Commissions (European Commission, 2025). This objective requires a very low carbon energy system, based on both renewable and nuclear energies, the two backbones of the future energy mix. The deployment of small modular nuclear reactors (henceforth, SMRs) across the EU could help countries reduce their dependence on fossil fuels, complement existing nuclear and renewable power plants, and contribute to Europe's future energy security. SMRs also offer several types of configurations and temperature ranges to create heat that can be used directly, or generate electricity, to offer mobility and decarbonise hard-to-abate sectors such as steel, cement, chemical, and refining.

This report leverages the Techno-Economic ecoSystem (TES) approach to map the worldwide SMR ecosystem, provides a first look at the global SMR industrial landscape and highlights EU strategic interdependencies, vulnerabilities, and opportunities⁴. It highlights where the EU has invested in SMRs and presents a network analysis of global SMR players to zoom in on innovation hubs and gatekeepers. This exploratory analysis could be used as a first step to further investigate supply chain dependencies, analyse them in relation to public perception and engagement, and provide strategic guidance to support government policies and international collaborations for SMR development and deployment.

Under today's global policies, world energy demand estimates suggest a 9% and 20% increase by 2030 and 2050 compared to 2023 levels, respectively (IEA, 2024b). However, these estimates might increase further depending on the pace of digitalisation, particularly in economies fostering data-driven AI tools⁵. Meanwhile, the International Atomic Energy Agency (IAEA) projects an increase in nuclear power between 40% and 250% by 2050. These numbers reflect the relevance of nuclear

¹ [EUR-Lex - 11957A/TXT - EN - EUR-Lex](#)

² [EUR-Lex - 12016E194 - EN - EUR-Lex](#)

³ See 'Energy production and imports: tables and figures', https://ec.europa.eu/eurostat/statistics-explained/index.php?title=Energy_production_and_imports#Source_data_for_tables_and_graphs

⁴ The JRC Technical Report by Righi et al. (2020) describes in detail the JRC TES methodology and the generation of the TES graph database. Fabiani et al. (2024) present a recent application of TES into the Advanced Manufacturing industry, adding new metrics and indicators by exploiting the information contained in the graph database.

⁵ See box 1.4 in IEA (2024b).

power to meet future energy demand. Out of this growth, SMRs should provide around one-fifth of the required additional power demand globally (IAEA, 2024a). The latest IAEA SMR “booklet” is cataloguing 68 SMR design and technology options worldwide, thereof 17 developed in Europe (IAEA, 2024b). Outside Europe, countries like China, Japan, Russia and India have already built several such reactors.

While energy consumption is expected to decrease in the EU under the current scenario –around 4% by 2030 and around 19% by 2050 compared to 2023–, efforts need to continue to meet policy objectives of the ‘Fit for 55’ package. Decarbonisation is one of the ambitious policy goals within the European Green Deal, with a 55% emissions reduction by 2030 and climate neutrality by 2050 (European Commission, 2024a). ‘Fit for 55’ is a set of legislative proposals designed to bring all sectors on track to reach these goals. Thus, it is a core component of the Green Deal.

Clean energy manufacturing must be an important contributor to decarbonisation. In 2023, the EU was the second largest installer of clean technologies in the world, behind China. SMRs could significantly contribute to increase the EU's clean energy production capacity. Nowadays, fifteen countries (Belgium, Bulgaria, Croatia, Czech Republic, Estonia, Finland, France, Hungary, Italy, The Netherlands, Poland, Romania, Slovakia, Slovenia and Sweden) have an active interest in SMRs⁶.

Drivers for the development and deployment of SMRs are also coming out in a growing number of countries outside the EU, targeting both electric and non-electric applications. The outcome of the first global stocktake under the Paris Agreement, brought nuclear energy back in the spotlight, as nuclear was declared as a low-emission technology (UNFCCC, 2023). More particularly, the interest in SMRs rests mainly in their claimed flexibility, feasibility and capability to combine with other energy sources, such as variable renewable energy. The reduced size of these reactors makes them in principle more affordable, thus reaching a wider range of users, from smaller countries to private companies. Their modularity would allow for on-site installation even in small and remote locations.

Moreover, SMRs incorporate advanced passive safety systems, which make them more appealing for users with less experience in nuclear power. The potential hazards of reactor meltdown with subsequent ecological and humanitarian consequences, issues with nuclear waste disposal, and fissile material diversion and proliferation have often led to criticism on nuclear energy fission. SMRs comprise scalable nuclear power plant designs that could reduce investment risks through incremental capacity expansion and proportional risk management as they contain substantially smaller radioactive inventory (Kessides and Kuznetsov, 2012).

The IEA estimates that SMRs could provide a total capacity of approximately 40 gigawatts (GWe) by 2050 in the most conservative scenario. In the most ambitious scenario, which sets out a pathway for the global energy sector to achieve net zero carbon dioxide (CO₂) emissions by 2050, there would be more than 1 500 reactors, with a total capacity of nearly 200 GWe (IEA, 2025b). The US, Canada, China, Japan, Russia and the UK currently adopt the most advanced SMR technology, although most of their projects are in the design stage of development. Many developing nations have also expressed their interest in this technology (IAEA, 2022). Indonesia, Saudi Arabia and

⁶ In May 2023, representatives from 14 EU Member States and the UK agreed to work together on a roadmap to deepen their cooperation and trigger the involvement of the EU in the field of nuclear energy, highlighting the need to foster innovation on SMRs and its capability to provide much of the additional electricity capacity needed by 2050. Italy has recently included SMRs in its national energy plans: https://www.mase.gov.it/sites/default/files/PNIEC_2024_revfin_01072024.pdf

South Africa are working on their gas-cooled reactor concepts. Brazil is designing its own pressurised water reactor concept, while Argentina is already constructing a reactor of the same type. India exhibits a dynamic SMR environment, with several operational units for its IPHWR-220 and ongoing designs of additional heavy water and sodium-cooled fast reactors (IAEA, 2025).

As a standard practice in the nuclear industry, SMRs must apply for licenses prior to deployment. The World Nuclear Association (WNA) acknowledges that this licensing process is a potential challenge to quickly scale up SMR installations, as design certification, construction and operation licence costs are not necessarily less complex or cheaper than for large reactors (World Nuclear Association, 2015). Other barriers to deployment include initial financing, security and safety procedures and an overall international regulatory framework (Hahn, 2023). For example, SMRs can be a cost-effective means of investing in hydrogen production. However, deploying and integrating SMRs into an independent energy market requires massive initial investments in Europe's infrastructure networks.

The European Commission established the European Industrial Alliance on SMRs⁷ and launched a call for membership on February 6th, 2024. Its objective is accelerating the development and deployment of SMRs by identifying the most promising, advanced, safe, and cost-effective SMR technologies. It also aims to strengthen an EU-wide supply chain, identify investment barriers, analyse funding opportunities, and explore new financial options for SMR development. Additionally, it seeks to identify future research needs for SMRs and advanced modular reactors, ensure a skilled workforce, and involve potential end-users such as energy-intensive industries, low-carbon hydrogen producers, and municipalities. Facilitating the rollout of SMRs will contribute to maintain the EU's global leadership in decarbonisation and energy transition. While remaining technologically neutral, the Commission recognises the potential of SMRs as an emerging low-carbon technology. However, Member States make their own decisions on whether to include SMRs in their national energy plans. In this regard, the Commission is committed to facilitating their safe development in Europe.

Box 1. What are Small Modular Reactors?⁸

According to the IAEA, small modular reactors are reactors incorporating advanced engineered features with a maximum output of 300 Megawatt electric (MWe) – while microreactors are any SMR under 10 MWe – and can produce up to 7.2 million kilowatt-hours (kWh) per day (IAEA, 2023b). While SMR is the generic term for these reactors, advanced modular reactors (AMRs) refer to designs based on non-light water technology. They all harness nuclear fission reactions to generate heat that can be used as-is or for the generation of electricity. The Generation IV International Forum (GIF) selected six concepts that target improved safety, sustainability, efficiency and costs (WNA, 2024a). The designs, which represent 'the future shape of nuclear energy', are the gas-cooled fast reactor, the lead-cooled fast reactor, the molten salt reactor, the sodium-cooled fast reactor, the supercritical-water-cooled reactor and the very high-temperature reactor.

Compared to conventional reactors, their small and modular features make SMRs and microreactors a potential flexible and tailored solution especially when it is not possible to use nuclear power reactors. Due to their smaller size, SMR developers claim that they need less space and less cooling water, while their

⁷ [European Industrial Alliance on Small Modular Reactors – European Commission](#)

⁸ The GIF official website provides a schematic view of Generation IV systems and their components. Access available at: <https://www.gen-4.org>

modular feature allows for mass production, reducing unitary costs through economies of scale. They are also deployable either as a single or multi-module plant. Small size and modularity also make SMRs easy to transport in trucks and installed on site. These reactors incorporate synergetic hybrid energy systems that supposedly make them combinable with alternative energy sources.

SMRs also incorporate advanced safety features and passive safety systems. Theoretically, these systems eliminate components prone to create failures and increase the reaction time allowed to operators in case of incidents or accidents. Unlike traditional reactors, SMRs do not need much engineering for safety including heat removal. In addition, since small reactors aim at replacing fossil fuel plants in many situations, the design of the emergency planning zone required is smaller.

By April 2025, IAEA's Advanced Reactor Information System (ARIS) reports 123 SMR designs and concepts globally from major technology lines, although the majority of them are in the design phase (IAEA, 2025). Interestingly, some reactors are claimed to be near-term deployable.

1.1 The strategic relevance of SMRs

The EU's new political agenda envisions a renovated single market that integrates critical sectors, such as digital, energy and finance (European Commission and Leyen, 2024). The lack of integration of these areas within the single market currently represents a major hurdle to European competitiveness (Letta, 2024). The Draghi report also hints at restructuring the common energy market as a critical element in accomplishing the necessary actions in the EU's roadmap for sustainable growth. In combination with other measures, these proposed changes will support the completion of three main actions: 1) closing innovation gaps with China and the US 2) decarbonising the economy; 3) and aiming to keep up with global competitiveness and reducing dependencies⁹. Nuclear energy, specifically SMRs, offers an opportunity to address these needs.

Today, the prosperity of societies is strongly associated with access to reliable and affordable energy, mainly in the form of heat, electricity and hydrogen. Energy enables effective food production; ensures clean water; provides lighting, heating, and cooling for homes and buildings; supports industrial processes; and powers transportation, communication and entertainment. As the world population continues to grow, human right institutions increasingly recognise access to energy as a fundamental right for the people.

Megatrend urbanisation also exacerbates the movement to higher energy consumption density. The extensive use of renewable energy alone has difficulties in keeping pace with this trend. Even in concert with energy efficiency measures, renewable sources cannot provide the high energy densities of fossil fuels and nuclear energy.

The digital revolution goes hand in hand with a dramatic rise in the energy needed to build, transport, deploy and power software and hardware systems. The International Energy Agency estimates that global data centres alone will need more than 800 terawatt hours of power by 2026 (IEA, 2024a). This is double the amount required by 2022. SMRs offer a means to address sustainability concerns that may undermine the productivity gains and benefits to social well-being of the green and digital twin transitions. With the acceleration of climate change, European citizens and politicians are becoming increasingly aware of the negative impacts of continued reliance on

⁹ See the Draghi Report (2024) for detailed analysis and sectoral recommendations on addressing the EU's innovation gaps. This report zooms in on what the EU can do to support innovation within the SMR ecosystem.

fossil fuels for a stable energy supply and as feedstock for essential products in the chemical and petrochemical industries. Moreover, the war in Ukraine is evidence that Europe's strong dependence on fossil fuel imports is weakening its economy, thus challenging its geopolitical weight. Europe cannot afford to overlook the geostrategic importance of building and maintaining critical energy intensive industries. Therefore, the urgency of implementation of solutions to limit the use of fossil fuel in Europe is becoming widely recognized and included in the objectives of the Net Zero Industry Act (NZIA)¹⁰.

To achieve these objectives, nuclear energy can play a part in securing access to affordable energy for the industry in countries that want to use it. While existing large-scale nuclear reactors have a reputation as a "cash cow" once written off producing substantial amounts of electricity at low cost, thus generating interest in Long-Term Operation¹¹. In contrast, nuclear new builds in Europe represent large strategic long-term infrastructure investments with an elevated risk of budget and schedule overruns, often linked to a lack of construction routines.

A report by the Massachusetts Institute of Technology (MIT) has discussed the interrelated technical, economic, environmental, and political challenges on nuclear energy to play a significant role in the future. SMR vendors have selected and integrated into their designs many of the recommendations highlighted in the report (MIT, 2018). As a next step towards deployment, these designs need to prove their feasibility to be licensed and built at scale. Such demonstration would corroborate the claim that the economics of replication can replace the current economics of size.

The European Commission has highlighted in various communications the strategic importance of SMRs. They can help to decarbonise the energy mix, increase power capacity in rural and remote areas, and stabilise energy grids with large fractions of variable renewable sources¹². The Commission acknowledges the potential of SMRs to achieve the energy and climate objectives of the EU Green Deal (European Commission, 2024a). More specifically, the Commission identifies nuclear energy as a power supply complement to renewable sources, which is critical to ensuring the EU's competitiveness (European Commission, 2024b). This vision has been recently endorsed in the Clean Industrial Deal, where one of the objectives is "support the acceleration of the development and deployment of small modular reactors (SMRs)" to boost clean energy demand (European Commission, 2025).

The strategic objectives mentioned above are not only important for Europe but also for countries interested in using nuclear power in their energy mix. Regarding international collaborative support for SMR development, demonstration and deployment, several initiatives deserve mention.

- The WNA provides a wealth of information regarding SMRs, including information about the policies and concepts developed by each country (WNA, 2024).

¹⁰ [EUR-Lex - 32024R1735 - EN - EUR-Lex](#)

¹¹ IAEA defines long-term operation (LTO) of nuclear power plants as the "operation beyond operation beyond an established time frame defined by the licence term, the original plant design, relevant standards or national regulations. Long-term operation should be justified by safety assessment [which] may take place within a broader regulatory process" (IAEA, 2018).

¹² Visit the European Commission website on Small Modular Reactors for a detailed overview on SMRs: https://energy.ec.europa.eu/topics/nuclear-energy/small-modular-reactors/small-modular-reactors-explained_en

- The International Atomic Energy Agency (IAEA) is running several working groups and Coordinated Research Projects on advanced reactors, including SMRs. This includes reactor-specific technical areas, non-electric applications and the integration of nuclear energy in Hybrid Energy Systems. The Advanced Reactor Information System (ARIS) database contains design descriptions of most evolutionary and innovative nuclear reactors (IAEA, 2025).
- The Generation IV International Forum (GIF) was created in 2001 as a co-operative international endeavour seeking to develop the research necessary to assess the feasibility and performance of fourth generation nuclear systems (Gen-IV systems), and to make them available for industrial deployment by 2030. Many of the currently proposed SMR concepts are of the Gen-IV type and significantly benefit from the available results, e.g. in structural materials development, characterisation, qualification and standardisation.
- The OECD-Nuclear Energy Agency (OECD-NEA) serves as technical secretariat to GIF. It also coordinates research into nuclear safety and regulatory frameworks. It also publishes regularly studies regarding the role of SMRs in future energy systems. It has produced several strategically relevant documents regarding SMR demonstration and deployment, and produced a dashboard with an evaluation of SMR concepts along a number of key criteria (NEA, 2024a).

At the European level, there are several activities relevant for SMR development and deployment in place:

- For more than two decades, the Euratom budget has financed many projects of relevance for SMR and AMR and their applications, with particular focus on safety performance¹³. This covers both the EU wide collaborative research effort via so-called “indirect actions” among Member State organisations and the “direct actions” of the JRC^{14,15}.
- Created in 2007 as the nuclear technology platform for the SET Plan, the Sustainable Nuclear Energy Technology Platform (SNETP) is a pan-European association uniting most relevant players in nuclear along a commonly agreed strategy from research to deployment¹⁶. All three of its pillars actively support SMR. SNETP organises regular events and acts as an interface between policymakers, research organizations, and industry.
- Most recently, in February 2024, the European Commission launched the European SMR Industrial Alliance¹⁷. The platform aims at facilitating and accelerating the development, demonstration, and deployment of the first SMR projects in Europe in the early 2030s.

1.2 The TES approach to map the SMR techno-economic ecosystem.

In view of recent developments in the SMR industry and the strategic importance of their development, there is a need to take stock of the current state of play at the global and EU levels.

¹³ https://research-and-innovation.ec.europa.eu/funding/funding-opportunities/funding-programmes-and-open-calls/horizon-europe/euratom-research-and-training-programme_en

¹⁴ <https://cordis.europa.eu/>

¹⁵ https://joint-research-centre.ec.europa.eu/jrc-science-and-knowledge-activities/small-modular-reactors_en

¹⁶ <https://snetp.eu/>

¹⁷ European Industrial Alliance on SMRs: https://single-market-economy.ec.europa.eu/industry/industrial-alliances/european-industrial-alliance-small-modular-reactors_en

By enhancing the understanding of the SMR ecosystem, this study aims at providing policy-relevant insights to ongoing discussions on the industry's state of play. To do so, the study highlights the predominant advanced reactor models in the global ecosystem and the most common technologies across leading players. The goal is to facilitate an informed reflection on how to use European resources for the deployment of SMRs. This report can also inform the European Industrial Alliance on SMRs by providing an overview of the SMR industry in terms of technologies, global competitors and the impact of EU funding programmes.

The JRC's Digital Economy Unit has developed and applied the TES analytical approach to map and explore the complex dynamics of industry's segments driven by rapidly evolving emerging technologies that are difficult to capture using traditional metrics or classifications. In this study, we apply the TES methodology to the quickly developing SMR industry, in order to be able to produce new metrics and insights at the global and EU level. This methodology relies on natural language processing methods and a system of keywords and queries used to define SMR activities, players and technologies (Calza et al., 2022). Keywords are curated by experts and extracted from relevant literature. The keyword list is provided in **Table 2** of Annex 1. To avoid ambiguity and potential false positives, search queries are often constraint by additional SMR-specific keywords, like "reactor", "advanced nuclear" or "generation IV". This is particularly relevant as nuclear technologies and processes can have applications in other industries, such as aerospace and defence.

Activities are identified by several micro-level data sources and categorized into three groups based on their nature and outcomes (De Prato et al., 2019; Righi et al., 2020):

- (a) Business activities are derived from companies' information on core operations, encompassing the production, supply or exchange of goods and services, or investments. Company data are obtained from multiple data sources (see Calza et al. 2022) for details).
- (b) Innovation activities consist of priority patent application filings obtained from EPO 's PATSTAT database¹⁸.
- (c) Research activities, referring to scientific publications in top journals, proxying frontier research. These activities are extracted from a set of publications from the Scopus repository¹⁹.

Players stand for entities that are engaging in the above-mentioned activities across the SMR ecosystem, in the sense that they participate in the design, development, deployment and commercialization of knowledge and products (goods or services) within the boundaries of SMR technologies. In identifying players, emphasis is placed on organizations rather than individuals, i.e. the applicant or developing organization in the context of patents, affiliation instead of individual authors in conference proceedings and publications, and always within the scope of SMRs.

An example of a player is a company that produces and/or deploys advanced modular reactors, like Westinghouse Electric Company. An example of an activity is a research publication on the cost-benefit analysis of the "modular construction" and SMR decommissioning (Mignacca and Locatelli, 2020).

¹⁸ <https://www.epo.org/en/searching-for-patents/business/patstat>

¹⁹ <https://www.elsevier.com/products/scopus>

Players are classified along with the following 3 types:

- (a) Companies;
- (b) Academic institutions and research centres;
- (c) Governmental authorities and bodies.

The TES mapping of the SMR ecosystem offers a global picture of research, innovation, and business activities related to SMR technologies from 2009 to 2023, and of the related actors. By merging several sources of micro data²⁰ into a graph database where the connections across datasets enrich and amplify the information contained in each individual source, thus proxying the structure of a complex ecosystem, the application of the TES approach offers a wealth of information on the global SMR landscape. The generated database enables the production of a set of indicators and metrics that provide a compelling characterisation of the key elements of the SMR ecosystem and of their interlinkages (Calza et al., 2022; Calza et al., 2023).

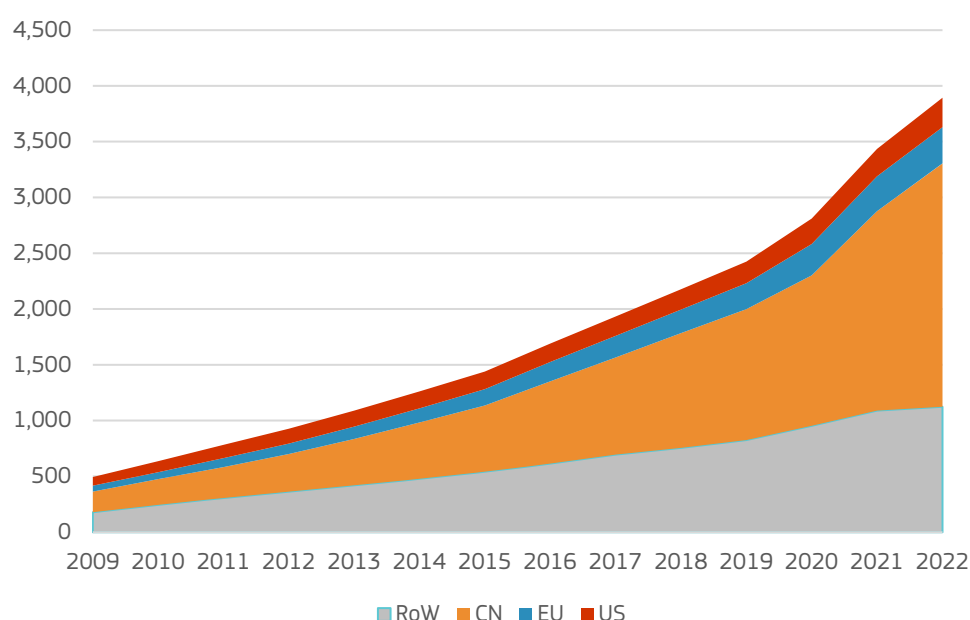
Therefore, the TES approach allows for several analytical avenues, to address policy-relevant questions regarding the design, diffusion and deployment of SMR technologies and components. The ultimate goal is to provide necessary evidence-based insights that support policy initiatives aimed at strengthening the competitiveness of the SMR industry.

²⁰ For the first study on SMRs, TES has mapped business activities on Orbis, Crunchbase, Dealroom and Dow Jones; research activities on Scopus; innovation activities on PATSTAT; and EU-funded projects on Horizon Europe, H2020 and FP7.

2 A global overview

The SMR ecosystem mapped using the TES methodology consists of 3 910 players participating in 9 255 activities. The mapping exercise captures the growing interest in SMRs over time. The growth in number of players involved in research, innovation and business activities has accelerated in recent years. **Figure 1** shows the evolution in terms of number of players by geographical area since 2009. The chart exhibits an uptick from 2020, reaching almost 4 000 players by 2022, mainly owing to the emergence of new Chinese players. China's dominance accounts for more than half – approximately 55%– of the total number of players in 2022. However, the gradual relevance of the SMR industry, as shown by the number of active players in the field, has become widespread across regions. According to IAEA data, only two SMRs were active by 2022: the Russian KLT-40S, a pressurised water reactor, and the Chinese HTR-PM, a high-temperature reactor. There are nine operational in 2024, with nearly a hundred more in the design or development stages. The renewed interest in nuclear energy strategic autonomy comes at a time of increasing geopolitical tensions.

Figure 1. Cumulative number of players by geographical area in the SMR ecosystem.



Note: Geographical areas refer to China (CN), European Union (EU), United States (US) and Rest of the World (RoW). Source: JRC TES database.

Private firms comprise a large part of the SMR landscape, with more than 85% of the total players. There are also more than 500 research institutes and universities, and 5 public institutions. In terms of activities, 94% of them relate to innovation and patenting, which is characteristic of an industry in the initial stages of development. Such a large share indicates players' interest in protecting SMR related technological innovations in view of industrial or market exploitation. Research and business activities account for approximately 3% of the total number of activities. Compared to other ecosystems mapped using the TES approach, such as the digital and the advanced manufacturing ones, SMR exhibits larger shares in innovation and research activities, and much lower shares in business activities. The SMR activity composition shows that the spotlight is still on obtaining commercial rights for concepts and technological developments prior to commercialisation. While research centres and universities participate in both research and innovation activities, government

institutions are seen uniquely partaking in research activities. This might be a sign of a lack of public-private partnerships on SMR projects worldwide.

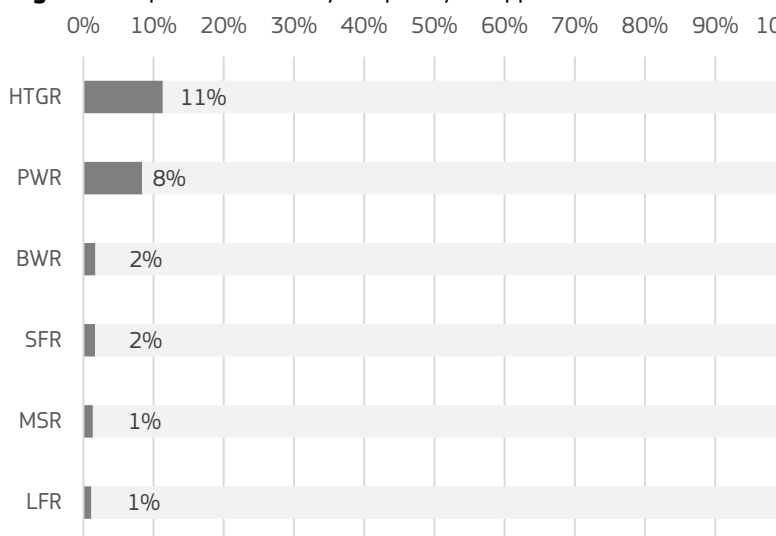
Compared to the global ecosystem, the EU presents a slightly different picture. The number of private firms is smaller, comprising less than 80% of the total number of players²¹. This share drops even further when EU-funded projects are considered in the analysis. Nonetheless, the share of EU business activities over the total (excluding EU-funded activities) is in line with the US and other Western countries, and larger than in Russia and Asian countries. The EU also dedicates more activities to research than the global average. As opposite, the share of innovation activities is below 60%. Except for the UK, which allocates 51% of its activities to innovation, all major competitors among private firms have above 75% of their activities on innovation²². The list includes Canada, China, Japan, Russia, South Korea and the US.

In terms of technologies, coolants are the technologies most frequently captured in the SMR ecosystem. This is not surprising, as the coolant system –whether using light water, gas, liquid metal or other elements as coolants– is an essential part of reactors to transfer heat away from the reactor core. Other technological aspects referring to the process of cooling off the reactor, like heat removal and demineralised water, a type of coolant, are also frequently targeted by firms' activities in the SMR ecosystem. These are also related to other important components of SMRs, such as pressure vessels and steam generators. Pressure vessels maintain the water at a high temperature, while steam generators convert it into steam. These are features of pressurised water reactors (PWR), the most common SMR design (**Figure 2**). Other SMR types are in the top-10 list of detected activities, among them high-temperature gas-cooled reactors (HTGR) and light-water reactors (LWR). All these types of reactors are bellwethers in the SMR field, as their design started back in the late 1940s.

²¹ See **Table 3** in Annex 2 for a list of main players by EU country.

²² While the EU stands out in research activities, the UK does so on business activities.

Figure 2. Top SMR models by frequency of appearance in TES database²³.



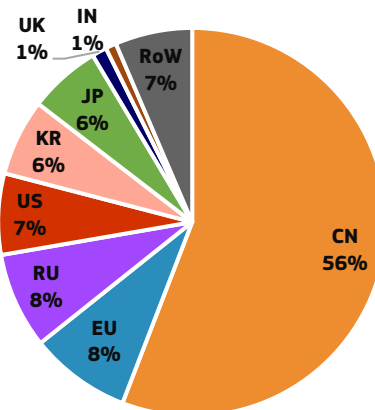
Note: Frequency based on number of activities that include an SMR model. Any activity, e.g., a patent, can refer to more than one technology (currently we do not distinguish a core business from side activities, but the presence of one or more keywords in the description of the activity is considered) Source: JRC TES database.

2.1 The global race for SMR technologies

The next step may be that of analysing the global network to understand the state of play in the SMR race. To do so we exploit the TES features, which builds on a network of players that cooperates in the design, manufacturing and deployment of SMRs. Similar to other digital and high-tech sectors, China's presence dominates (**Figure 3**). Chinese players represent almost 60% of the total players in the SMR ecosystem, and their dominance is evident in innovation. It has by far the largest share of firms, with nearly 2 000 of them taking part in SMR activities. It also has the largest number of research centres, but its share at the global level decreases to 40%. In addition to China, other major nuclear competitors evenly distribute participation in the SMR ecosystem. The EU slightly leads ahead of Russia, the US, South Korea and Japan. However, the US hosts several leading SMR firms, such as NuScale Power Corporation, Holtec International and Westinghouse Electric Company.

²³ High-temperature gas-cooled reactor (HTGR); pressurised heavy-water reactor (PWR); boiling water reactor (BWR); sodium-cooled Fast Reactor (SFR); molten salt reactor (MSR); lead-cooled Fast Reactor (LFR). See List of Abbreviations for the definition of other acronyms in this report.

Figure 3. Share of players by main geographical areas on TES SMR ecosystem.



Note: Geographical areas refer to China (CN), European Union (EU), Russia (RU), United States (US), South Korea (KR), Japan (JP), United Kingdom (UK) and India (IN). Source JRC TES database.

Similar to the breakdown by type of player, nearly 85% of the activities relate to patenting. Firms are the leading actors in patenting initiatives, with only business participation in nearly 80% of them. Research centres generate 16% of patents, while less than 4% are the result of business and research partnerships. China dominates innovation activities, partly favoured by less demanding eligibility criteria and subsidised industrial patenting systems. The share of innovation activities decreases to 71% when excluding Chinese players. Russia takes second place in innovation activities, closely followed by South Korea, Japan and the EU. Surprisingly, the US ranks sixth in terms of the number of patents.

In terms of commercial development, there are no significant differences between the main competitors. Accounting only for business activities, the US has the highest share with nearly 25% of activities, followed by the EU and China, each of which absorbs around 20% of global commercial activities. The UK, Canada, South Korea, Japan and Russia split much of the remaining market shares. Notwithstanding, the UK has the highest allocation to business activities of the total number of activities registered in the country.

The EU has a well-developed research network. The EU not only leads in the number of research activities but has nearly double the research institutions and universities involved in SMR activities compared to Russia, the US and South Korea, and even more so compared to other countries. We identify 67 research institutions and universities in the EU, behind the 218 of China.

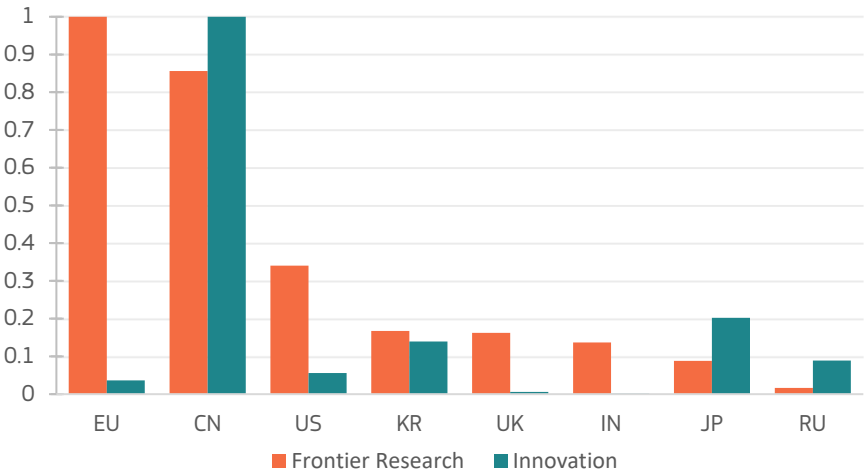
A relevant feature of EU research players is their scientific leadership. This is reflected in the total number of research activities, where the EU dominates with 28% of the global scientific volume.

Figure 4 may also highlight the capacity of EU actors to generate independent SMR research²⁴. The Independence ranking reflects if a country generates activity –in the case of research, scientific

²⁴ The Independence Ranking is a score regarding the countries' generation of activities without the participation of foreign players. The participation in each activity is weighted according to the fractional counting of the same activity. The fractional counting equals to $1/(n. \text{ of players involved})$. Finally, the sum of the fractional counting of the activities is computed for each player.

publications- with or without the participation of foreign players. The lower the Independence ranking, the higher the number of collaborations with foreign players. The EU has a value of nearly one, which means that local experts produce most of the scientific research on SMRs²⁵. This contrasts with other countries, where scientific publications partially rely on foreign researchers. However, the EU generates less independent innovation –i.e., without the participation of non-EU players– than most of its main competitors. The EU’s challenge is thus transforming its domestic research excellence into innovation and expanding its business network²⁶.

Figure 4. Relative independence ranking by type of activity, excluding EU funded projects.



Note: Geographical areas refer to European Union (EU), China (CN), United States (US), South Korea (KR), United Kingdom (UK), India (IN), Japan (JP) and Russia (RU). Source: JRC TES database.

Two thirds of the collaborations in research activities are across Member States. Collaborative activities in R&I expand significantly, both with EU and non-EU players, when we account for EU-funded activities. Many of the external EU collaborations are with the UK and other European nations.

In the global SMR ecosystem, three out of four R&I activities are non-collaborative. Collaborative activities represent half of the global research output, while the ratio declines to one fourth in the case of innovation activities²⁷. The collaboration can be either internal –across local players– or external –across players from different regions–.

It is interesting to examine the technologies that the most active players are aiming at. **Table 1** presents the players that are associated with the highest number of different activities, along with the most pervasive SMR technology in their activities. The most frequently tracked technologies are in line with those presented in **Figure 2**. Pressure vessels are widely used in the activities of leading

²⁵ In this ranking, the EU is considered as a bloc. That is, if all participants in each activity are from EU countries, it is considered as a fully independent EU activity.

²⁶ While the EU is well positioned in terms of business activities, the share of firms over total players is, along with the UK, the lowest among major competitors.

²⁷ One (non-collaborative) or more than one (collaborative) player can conduct innovation and research activities.

players. In addition, pressurised water reactors (PWR) and high-temperature gas-cooled reactors (HTGR) are the reactors of choice among industry leaders.

Table 1. Top-10 players and the most repeated keywords associated with them.

Player	Technology keyword
Nuclear Power Institute of China	PWR
China General Nuclear Power Corporation (CGN Power Co.)	Pressure vessels
Tsinghua University	HTGR
China Nuclear Power Engineering Co., Ltd. (CNNC)	PWR
Korea Atomic Energy Research Institute (KAERI)	Nuclear fuel
Toshiba Corp	Pressure vessels
Xi'an Jiaotong University	PWR
GE Hitachi Nuclear Energy (GEH)	Pressure vessels
Xi'an Thermal Power Research Institute Co., Ltd.	HTGR
Korea Hydro & Nuclear Power Co., Ltd. (KHNP)	Steam generators

Source: JRC TES database.

2.2 The strategic position of the EU on Small Modular Reactors

One of the main challenges the EU faces for a smooth development of the SMR industry is the lack of unanimous consensus. This is reflected in their National Energy and Climate Plans (NECPs) and in the current differences in deployment of nuclear infrastructure. Germany, Greece, Ireland, Italy and Austria have no operating nuclear reactors. Germany has historically opposed nuclear energy for a long time and closed its three remaining nuclear plants in April 2023²⁸. Italy had banned nuclear power plants in 2011 after a referendum, but it is experiencing renewed interest. Austria banned the construction of nuclear power plants much earlier, in 1978, while Spain still has active plans for a nuclear phase-out. On 15 May 2025, the Belgian Chamber of Representatives approved the amendment to the Nuclear Phase-Out Law. However, European public opinion and certain governments seem to be changing their positions. The Project Europe Research poll in nuclear energy shows that EU citizens' opinion about nuclear energy has significantly improved since 2022²⁹.

The TES database maps the number of players and business, innovation and research activities in the EU's SMR ecosystem, reflecting Member States' interest in the field based on their participation in these SMR-related activities over time. **Figure 5** shows the evolution of the SMR ecosystem across the EU based on the number of players participating in their activities. The chart excludes EU-funded projects, to capture the countries' interests in the absence of European public support.

²⁸ The recently appointed German government has signalled intentions to revert its opposition to nuclear power.

²⁹ *Public support for nuclear energy in Europe is growing*, Századvég Foundation (2023), <https://szazadveg.hu/en/cikkek/public-support-for-nuclear-energy-in-europe-is-growing/>

Over this ten-year timespan, Czechia, Finland, Hungary and Slovenia have experienced larger ecosystem growth than other Member States with a similar SMR landscape in 2012. Interestingly, countries with a historical background on nuclear energy that are decommissioning power plants have also experienced a rapid growth in the number of SMR players. This signals the capacity of firms and research centres to thrive independently of the national political support for nuclear energy generation. The network shows how players from Germany and Spain provide a relevant contribution to the European SMR ecosystem.

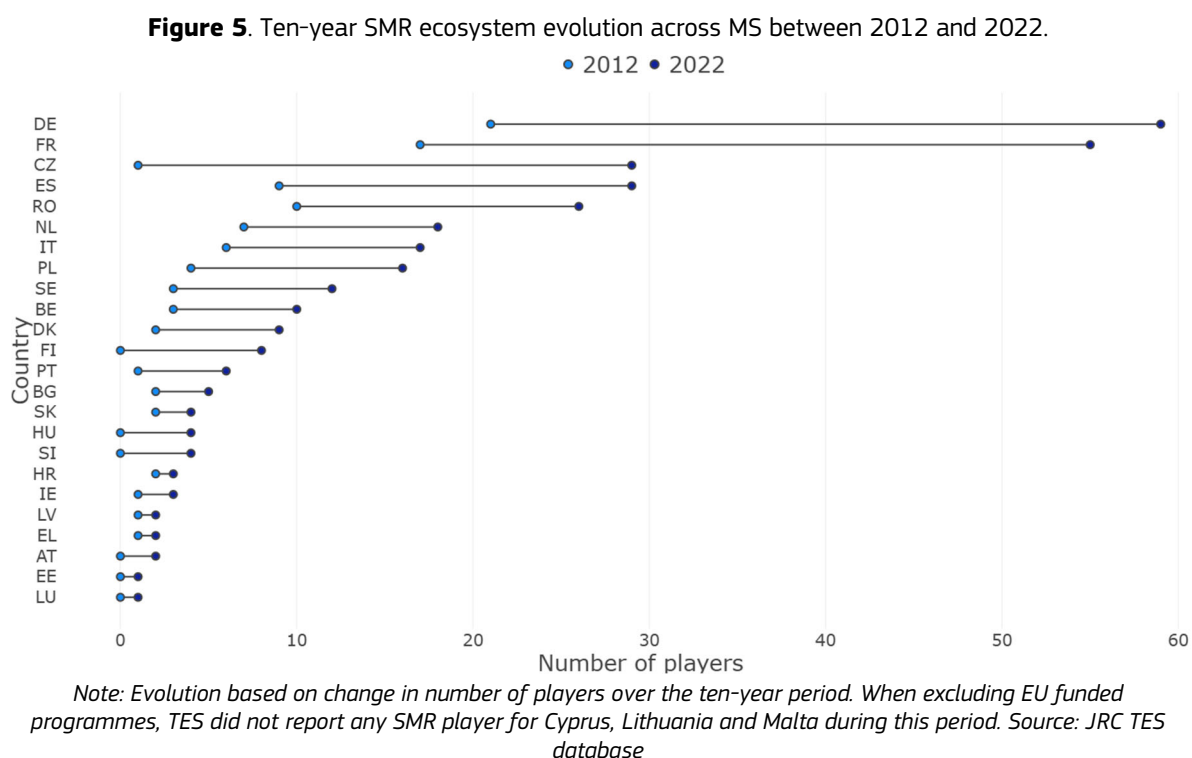
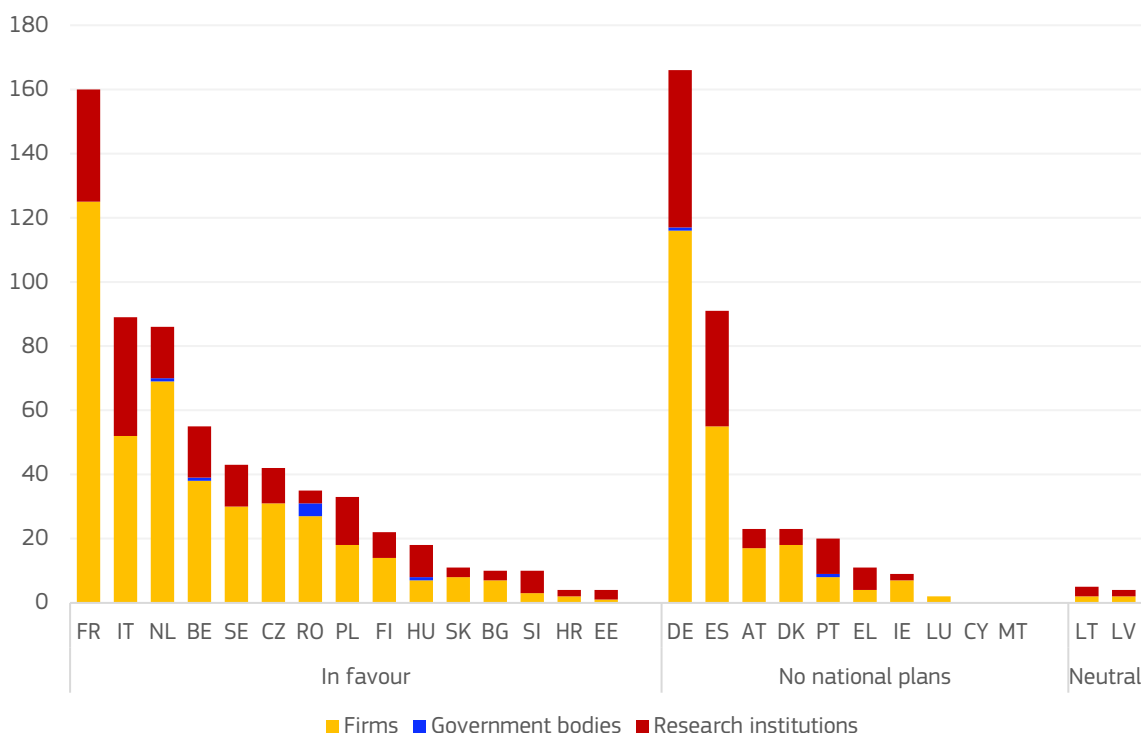


Figure 6 shows the Member States' player distribution across firms, government bodies and research centres institutions by 2022. The bar chart includes data on EU-funded projects, thus showing the full size of the SMR ecosystem when counting for EU public support. The chart also groups countries based on their views on nuclear power. Player composition is roughly similar across EU countries, with the largest share of players being private companies, followed by research institutions. Romania has the greatest participation of government bodies in SMR activities.

Overall, EU Member States (MS) with a proactive approach to nuclear energy tend to show a larger SMR ecosystem than those with a reluctant view, with the exception of Germany and Spain. These two countries have a background in nuclear production and research, and many of their players continue to engage in SMRs. However, the low rate of participation of countries opposing nuclear energy signals potential barriers to the full-fledged development of the EU SMR ecosystem. Overcoming this public support fragmentation would help to streamline the design and deployment of SMRs.

Figure 6. SMR players in EU countries by type and stance on SMRs.



Note: It includes players found in EU-funded programmes. "In favour" refers to countries publicly showing interest in SMRs; "No national plans" refers to countries leaving nuclear out of their energy strategic plans or those without operating nuclear reactors and not publicly showing interest; "Neutral" refers to countries assessing the possibility to include nuclear in their strategic plans. Source: JRC TES database^{30,31}.

The chart also hints at some correlation between the GDP size and the SMR ecosystem size when including EU-supported programmes. For example, the inclusion of EU-funded projects dilutes the weights of Czechia and Romania in favour of the largest economies that host larger R&I networks. EU support should make sure to foster initiatives across countries and players in the booming SMR environment.

2.2.1 The impact of EU-funded programmes on the development of SMRs

The analysis of activities from the TES database provides additional insights. First, it highlights the ubiquitous presence of EU-funded nuclear related projects in Member States' SMR network. **Figure**

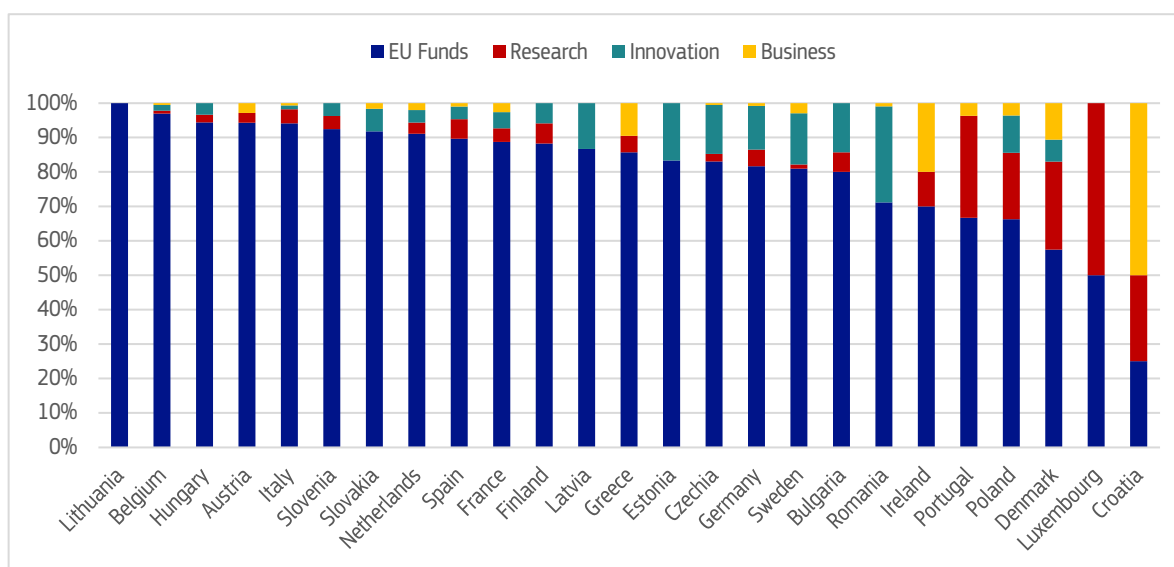
³⁰ During the World Climate Action Summit of the 28th Conference of the Parties to the UN Framework Convention on Climate Change held in December 2023, 25 countries, including 12 Member States, committed to triple nuclear energy capacity by 2050. These countries are Bulgaria, Croatia, Czech Republic, Finland, France, Hungary, The Netherlands, Poland, Romania, Slovakia, Slovenia and Sweden. These countries are part of the 'nuclear alliance', a coalition of Member States around the idea of nuclear power being a way to ensure simultaneously energy sovereignty and decarbonisation and to boost the economy. Belgium and Italy participate as observers.

³¹ Latvia and Lithuania are considering including nuclear power, and more particularly SMRs, as part of their national energy strategies.

7 represents the weight of EU-funded actions in supporting the activities of Member States. EU programmes trigger activities even in nuclear-sceptic countries. This is evident in the TES network, where collaboration in EU-funded research initiatives reveals the presence of players from countries that are less interested in nuclear energy. This is the case for Austria and Spain, with 94% and 89% of their activities extracted from EU-funded programmes, respectively. Additionally, as the relevant ecosystem is analysed, their participation might not be specific to SMR development, but to technologies or techniques with use cases beyond the industry. For example, almost half of Austrian activities are associated with the keyword “Technology readiness levels (TRL)”. This methodology estimates the maturity of technologies and it has applications in various fields, including defence and space (Héder, 2017). In Germany, the most repeated keyword is also TRL. In Denmark, while EU-supported projects do not relatively weigh as much as in other Member States, most of the remaining players are research centres and universities participating in scientific publications³².

While the chart emphasises the importance of EU projects, it also shows some heterogeneity across countries in composition of other activities. For example, the EU supports all activities registered in Lithuania, while these activities represent just one-fourth of the total in Croatia. There are differences beyond the activities of EU programmes. In some countries, business networks actively participate in innovation and patenting, while others focus on commercialisation. When excluding EU-supported R&D activities, France and the Netherlands show an SMR ecosystem with equally distributed activities on research, innovation and business activities.

Figure 7. Composition of EU countries' activities by type.



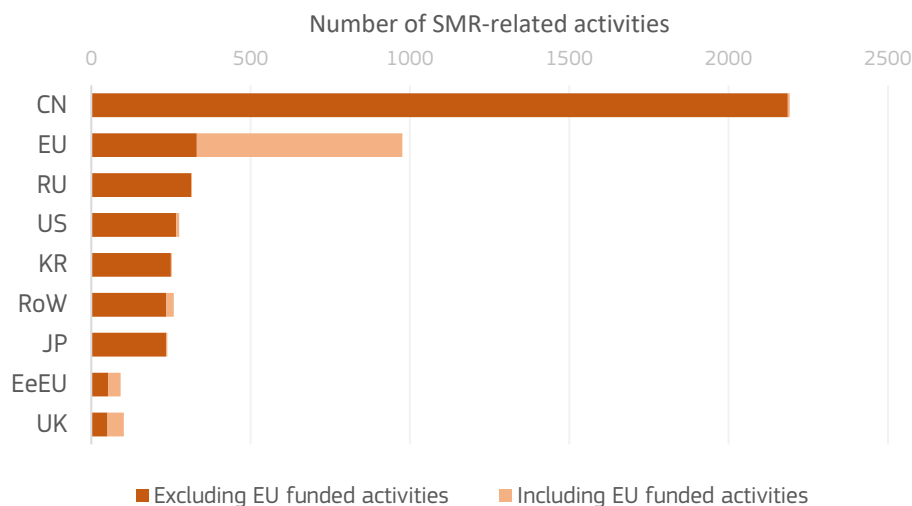
Source: JRC TES database.

EU projects are also crucial to competitiveness. **Figure 8** shows the size of the EU SMR ecosystem with and without public support and compares it to other geographical areas. The size of the

³² See **Table 2** in Annex 2 for a summary of the main players by Member State.

European SMR ecosystem, measured by the number of SMR activities, is more than double when EU projects are included.

Figure 8. Comparative weight of EU in the global landscape with and without EU-funded programmes.



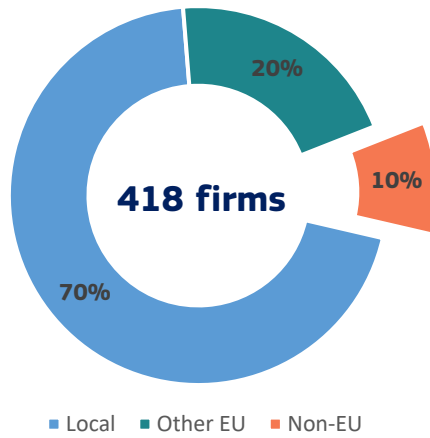
Note: Geographical codes are China (CN), European Union (EU), United States (US), South Korea (KR), Japan (JP), United Kingdom (UK), Europe excluding the EU (EeEU), Russia (RU) and rest of the world (RoW). Source: JRC TES database.

2.2.2 Strategic investment on EU firms

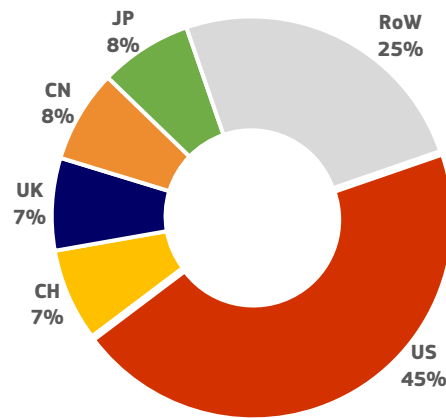
Zooming into business activities, this section explores the state of play of SMR firm ownership in the EU. Local investors own 70% of SMR firms, whereas EU investors control 20% of firms registered in another Member State (**Figure 9**). This indicates a strategically positive local control of the overall SMR business network. Focusing on the owners' country of origin for the 40 firms controlled by third countries, American capital dominates. US investors control nearly half of these firms, while the UK, Norway and Switzerland together account for another 20%. China controls only three EU firms.

Figure 9. Ownership of EU firms in the SMR ecosystem.

a. Ownership distribution of EU firms.



b. Country of origin of foreign-owned firms in the EU.

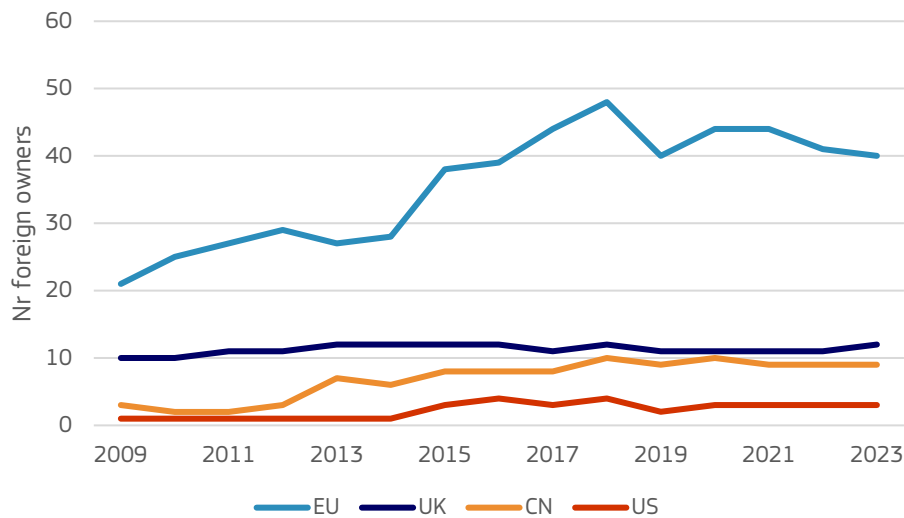


Note: Foreign ownership defined as the global ultimate owner holding more than 50% of shares. Geographical codes are China (CN), United States (US), Japan (JP), United Kingdom (UK), Switzerland (CH) and rest of the world (RoW). Source: JRC TES database.

The time series reflects that the share of foreign-owned firms has historically been higher in the EU compared to other competing countries. This gap has increased over time, peaking in 2018. However, that year, there was a tipping point. Since then, the share of foreign-owned firms has stabilized. At the same time, the overall number of EU firms has increased, thus foreign-owned firms now represent a lower share of the European SMR business network. These changes in ownership dynamics reflect the strategic importance of SMRs: not only are firms entering the locally owned SMR ecosystem, but also some foreign-owned firms pass into the hands of EU investors. **Figure 10** shows this trend leaving out EU firms with an owner from a different EU country, as we assume these are domestically-owned firms at the EU level³³.

³³ Note that **Figure 9** and **Figure 10** show a subsample of EU firms, as there is no ownership for the totality of firms captured in the TES network.

Figure 10. Number of foreign-owned SMR firms by geographical area over time.



Note: Foreign ownership defined as the global ultimate owner holding more than 50% of shares. A firm located in the EU is foreign when the owner is outside the EU. EU firms controlled by owner in other EU Member State are domestic. Geographical codes are European Union (EU), China (CN), United States (US) and United Kingdom (UK), Source: JRC TES database.

2.3 Network Analysis of the SMR ecosystem

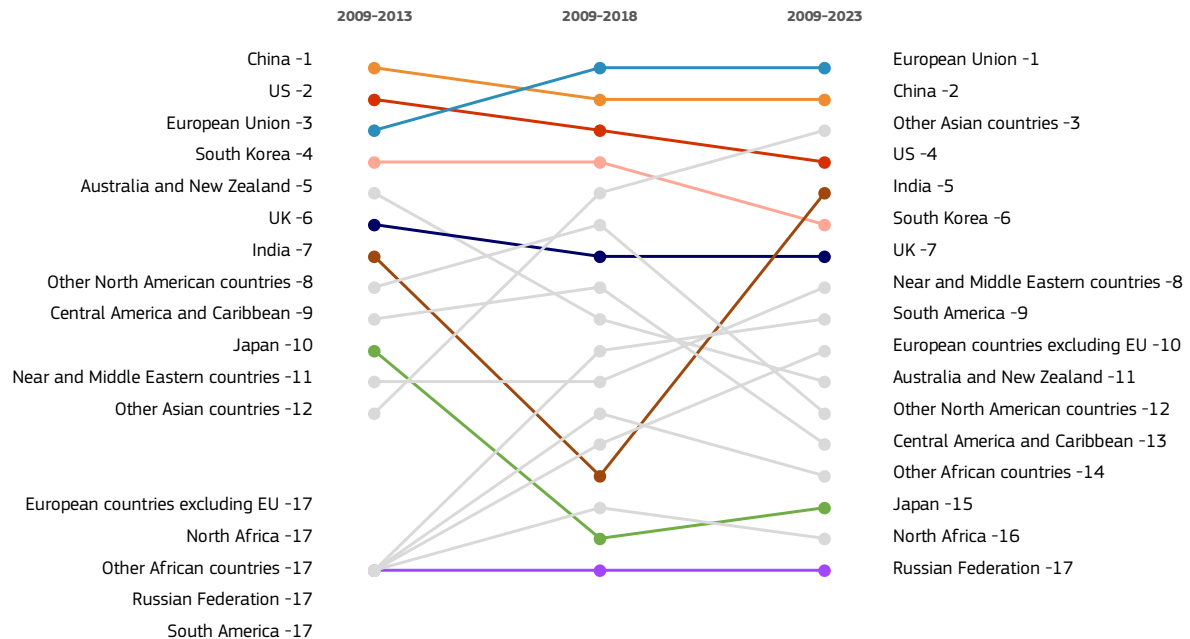
The granularity of the microdata used to generate the database of players and activities in the SMR ecosystem allows for the identification of specific groups of players or entities and assigns them to roles and positions. These assignments let the TES approach build a representation of the global network (Calza et al., 2022). Nodes, aggregated by geographical area, give shape to the SMR network when considering their collaborations in research and innovation activities – that is, co-authorship of scientific publications or co-patenting across geographical areas³⁴. By forming a web of collaborative research and innovation activities, these explicit and formal interconnections depict the SMR ecosystem as a network of interdependent entities connected by a flow of knowledge.

The network projection thus delineates the features and structure of the SMR ecosystem as a whole or for specific groups, and at various levels of aggregation, such as individuals or geographical area. The analysis of the structure of the network can reveal how knowledge flows through and across its parts, detecting whether certain nodes occupy a particularly relevant and strategic position (i.e. bottlenecks, gatekeepers) and whether this may reveal the existence of important relations and connections from the perspective of strategic autonomy.

³⁴ The SMR network projection is a (monopartite) network of players and collaborative activities in the SMR ecosystem and follows this structure: i) nodes are groups of players aggregated by geographical area; ii) edges represent collaborations in research and innovation activities across nodes – thus, across players from different geographical areas. Nodes and edges can take different forms, depending on the chosen attributes –e.g., specific categories or technological domains–, other than geographical categorization.

In network analysis, the centrality indicator provides a proxy for the importance of a node in terms of its contribution to the overall level of activities in the network (connectedness). **Figure 11** presents the ranking of research centrality by geographical area. Thus, the indicator accounts for each node as the number of collaborative research activities in a given period. The chart also shows how centrality evolved over time in each node, comparing the three different periods.

Figure 11. Centrality ranking: connectedness in the network of research activities.

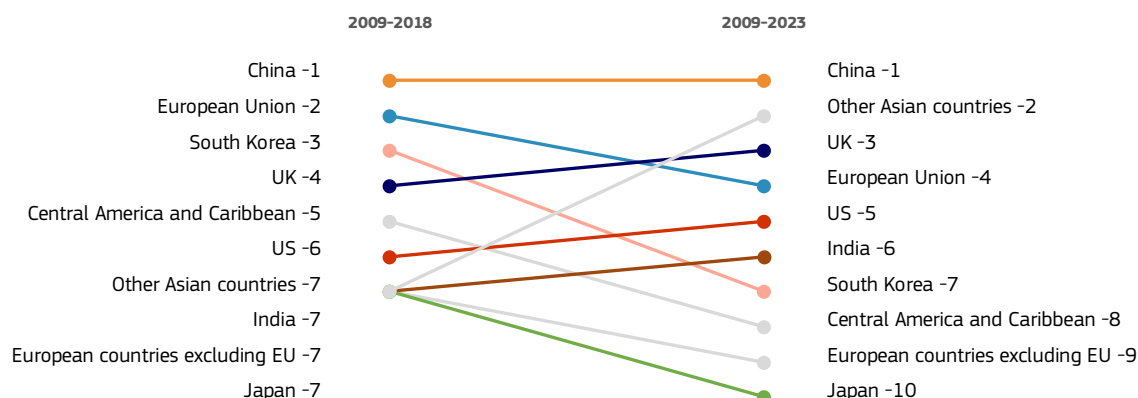


Source: JRC TES database.

The EU leads the ranking in 2023, followed by China and other Asian countries –mainly Pakistan, Malaysia and Taiwan– and the US. The EU has moved from third to first position in the ranking, while the US has fallen from second to fourth as its focus is on the deployment and commercialisation of SMR models. Other European countries also improved their rankings between 2013 and 2023. Interestingly, Russia remains at the bottom of the ranking throughout the period, despite being one of the first countries to have operational SMRs. It is difficult to determine whether Russia relies uniquely on national research, their findings are not publicly available, or the development of their SMR designs requires a lower amount of background research. The EU did not have operational SMRs by 2023, but its score in the research centrality ranking confirms the existence of a valuable scientific network as a sound basis for design and development, especially for AMR.

Next, the TES approach can identify players that act as “bridges” and “control points.” The indicator for betweenness centrality goes a step forward by measuring the influence and control that a node can exert on the rest of the SMR ecosystem (strategic position). In other words, this indicator proxies the ability of a node to influence the rest of the network by acting as an intermediary. This is strategic in two ways. First, a node with high betweenness centrality acts as a bridge connecting non-otherwise connected players in the network whose structure would be changed and disrupted without an intermediary node. Second, this node also acts as a bottleneck, as it can exert a relevant influence on the rest of the network and can steer, control or even interrupt the knowledge flowing to and from other nodes (Righi et al., 2021; Samoilu et al., 2020).

Figure 12. Betweenness ranking: strategic position to control research activities.

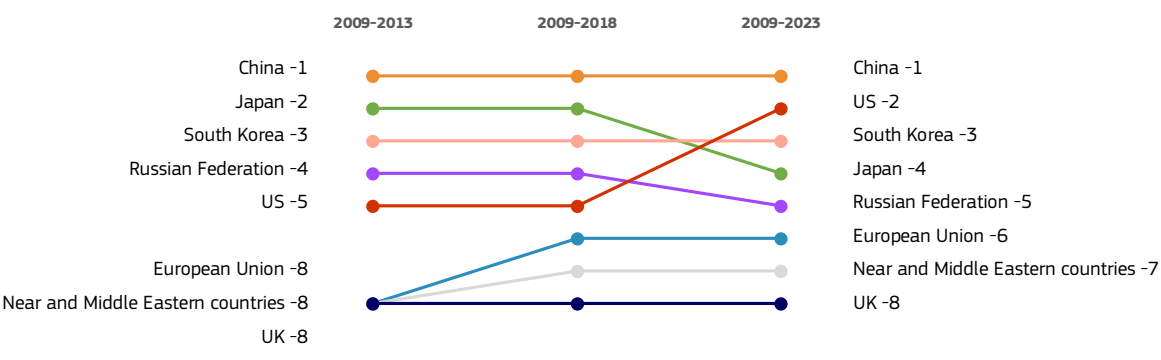


Source: JRC TES database.

Figure 12 presents the betweenness ranking of the scientific network. Despite performing a substantial volume of research activities, the strategic position of the EU is not as influential as expected based on previous metrics. Moreover, the ranking position has worsened in 2023 compared to 2018. China leads this indicator, followed by other Asian countries. The good performance of players in Pakistan, Malaysia and Taiwan led to this high ranking. The UK is in third place, while the US and India have improved their strategic positions by 2023. Despite its top position in terms of the number of research activities and collaborations in research, the EU's lower betweenness ranking reveals the need to reverse the downtrend by strengthening international scientific collaboration with leading institutions.

Figure 13 shows the ranking of geographical areas based on the betweenness indicator for innovation. China exerts a dominant influence on patenting activities, revealing a solid capacity to control the flow of information across separated nodes in the network. The US has improved its strategic position, becoming second in ranking, whereas the Russia ranks fifth in 2023. Despite having moved up from the last position, the EU remains near the bottom of the ranking. This reveals the scarce influence and control that the EU can exert on innovation knowledge flows across the SMR network. The low score of the EU in the betweenness indicator for innovation indicates a lack of strategic partnerships in these activities. Furthermore, the EU is present in less than 3% of innovation activities when excluding EU-supported programmes.

Figure 13. Betweenness ranking: strategic position to control innovation activities.



Source: JRC TES database.

3 The influence of SMRs on industrial ecosystems

SMRs can contribute to the development of other industrial ecosystems by providing a stable supply of energy. Looking at the industrial composition of SMRs is a useful and policy-relevant exercise, as it can help design better policies for the EU's industrial competitiveness. The presence of SMR activities offers a raw indication of the “critical mass” of SMR capabilities in a given industrial ecosystem. In this section, we crosscheck the data on SMRs with those of each industrial ecosystem, to identify the level of integration across them³⁵.

This analysis reveals the specialised nature of SMRs. Only 15% of SMR activities are strictly linked to others belonging to one of the industrial ecosystems. In other words, only a small percentage of SMR's activities are related or applied to an industrial domain different from the strictly considered SMR one. **Figure 14** shows the distribution of SMR activities showing an overlap with activities in industrial ecosystems. As expected, SMR activities mostly fall within the industrial ecosystems associated with power generation.

‘Energy-intensive Industries’ is the most concurrent industrial ecosystem with 43% of overlapping activities. SMRs are reliable tools in energy-intensive industries. This is also the case for digital and agri-food ecosystems. Growing energy demands from data centres are a major opportunity for SMRs (see sub-3.2.32). Companies such as NuScale are collaborating with energy solution firms to explore the use of SMRs as a reliable, low-carbon power source for data centres. The agri-food ecosystem also relies on energy-intensive operations. This is the case for food processing, which requires energy for heating, cooling, and electricity. Drying agricultural produce uses process-heating systems like heat pumps and desalination systems used for irrigation in water-scarce areas. Such processes require vast amounts of energy.

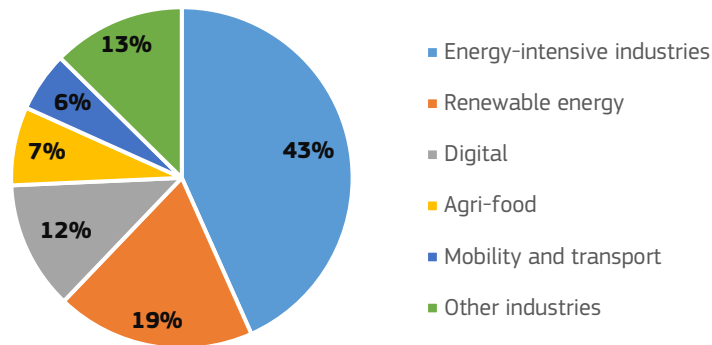
To reduce energy dependency and ensure decarbonisation, the EU also needs to increase hydrogen production at the required large (Mt/year) scale. Hydrogen is particularly valuable as both feedstock and reactant to decarbonise hard-to-abate sectors in heavy industries (steel, cement, chemistry, and fertilisers) and transportation (aviation and maritime transport). A recent study demonstrated the cost-effectiveness of SMRs in hydrogen production³⁶. However, deploying and integrating SMRs into an independent energy market requires massive initial investment in Europe's infrastructure networks.

The second-largest overlap is with the ‘Renewable Energy’ ecosystem. This ecosystem includes hydrogen production, where experts believe that small and microreactors can be a critical source of power supply. In addition, SMR developers claim that their models can be plugged together with renewable energy sources in energy systems. This feature would allow SMRs to function as complementary components in the transition to a low-carbon economy. Thus, SMRs can theoretically provide stability and resilience to the energy system during the twin transition, where the energy-intensive digital transformation relies on hybrid energy systems. That is, when a renewable energy supply is paired with small nuclear reactors.

³⁵ See European Commission (2023) for definitions and detailed discussion of industrial ecosystems.

³⁶ *SMRs cost-effective in hydrogen production, study finds.* <https://world-nuclear-news.org/Articles/SMRs-cost-effective-in-hydrogen-production,-study>

Figure 14. Engagement of industrial ecosystems and SMR technologies.



Source: JRC TES database.

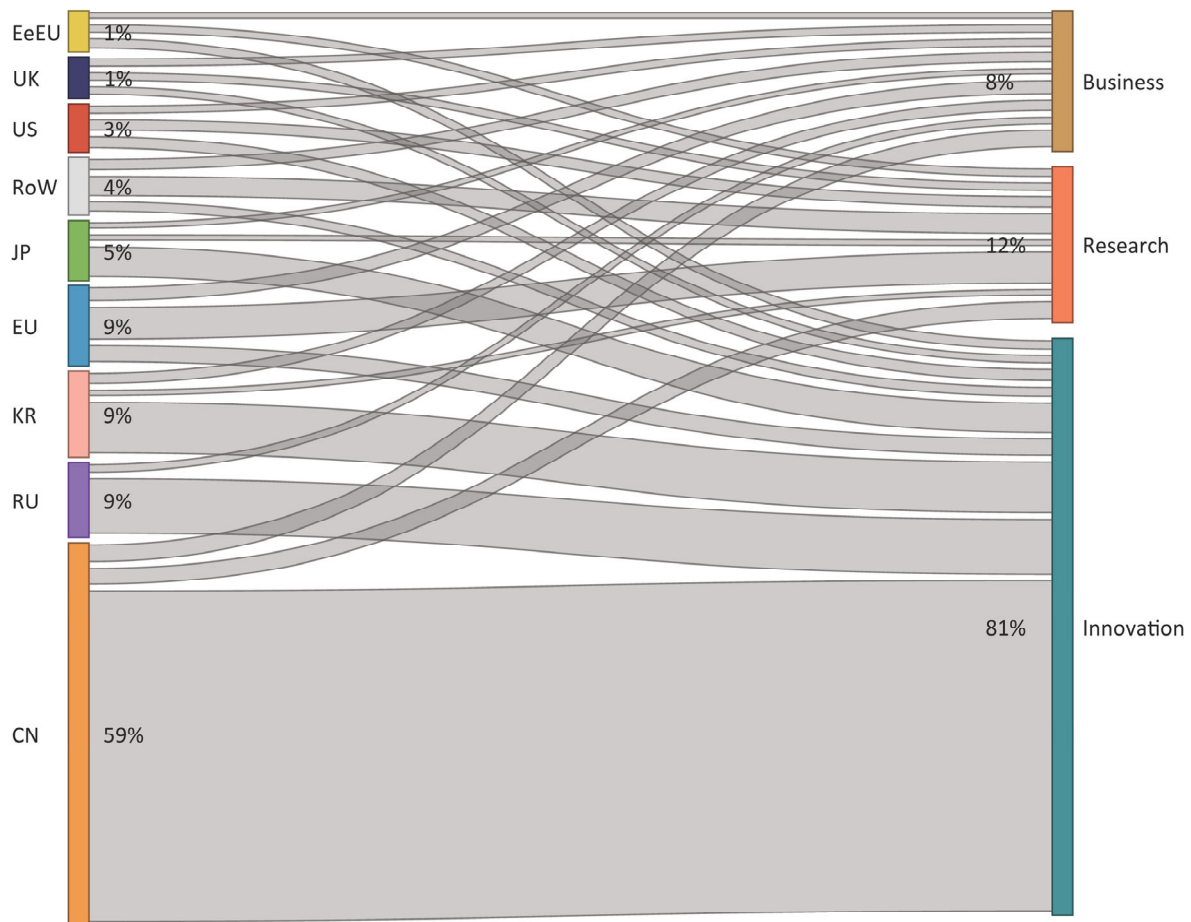
3.1 A relevant supplier for energy-intensive industries

European Member States are increasingly interested in maintaining their existing critical energy-intensive industries (EIs) infrastructure for geostrategic reasons. Industries such as steel, chemistry, cement, fertilisers, pharmaceutical products can effectively help countering the disadvantages of globalisation, thus granting a higher degree of strategic technology autonomy. Furthermore, competitiveness in these industries can prevent the loss of large investments to countries outside the EU and the relocation of industry branches to regions with lower energy costs. These regions also tend to have less restrictive emissions policies, and thus relocation can lead to carbon leakage. The EU could thus preserve skills and jobs within its borders, and the industry could continue to produce according to European standards and ethics (e.g., avoidance of child labour).

In the following analytical exercise, we identify activities that fall concurrently within the domains of EIs and SMRs (**Figure 15**). From the activities, we extract the players –firms, research centres or government institutions– participating in them and the registered location. China accounts for 59% of the identified activities, most of them related to patenting. Like the SMR ecosystem, the activity composition in the overlap with EIs is dominated by innovation with 81% of the activities. Russia, Japan and South Korea also dedicate most EIs activities to innovation. The captured business activities in this domain are, in broad terms, equally distributed across regions.

Except for China, the EU shows a superior performance compared to other competitors. However, it struggles to convert research into innovation. It is the only region with the largest share of activities allocated into research. At the country level, Germany and France lead in number of EI-SMR overlap. Germany has more research institutes, while France has more firms than any other EU country. French firms working at the junction with EIs includes Cyclife Digital Solutions –formerly Oreka Solutions–, a subsidiary of the EDF Group.

Figure 15. Geographical distribution of players in the ELLs-digital field and their activities.



Note: Geographical areas refer to China (CN), Russia (RU), South Korea (KR), European Union (EU), Japan (JP), rest of the world (RoW), United States (US), United Kingdom (UK), and Europe excluding the EU (EeEU). Source: JRC TES database.

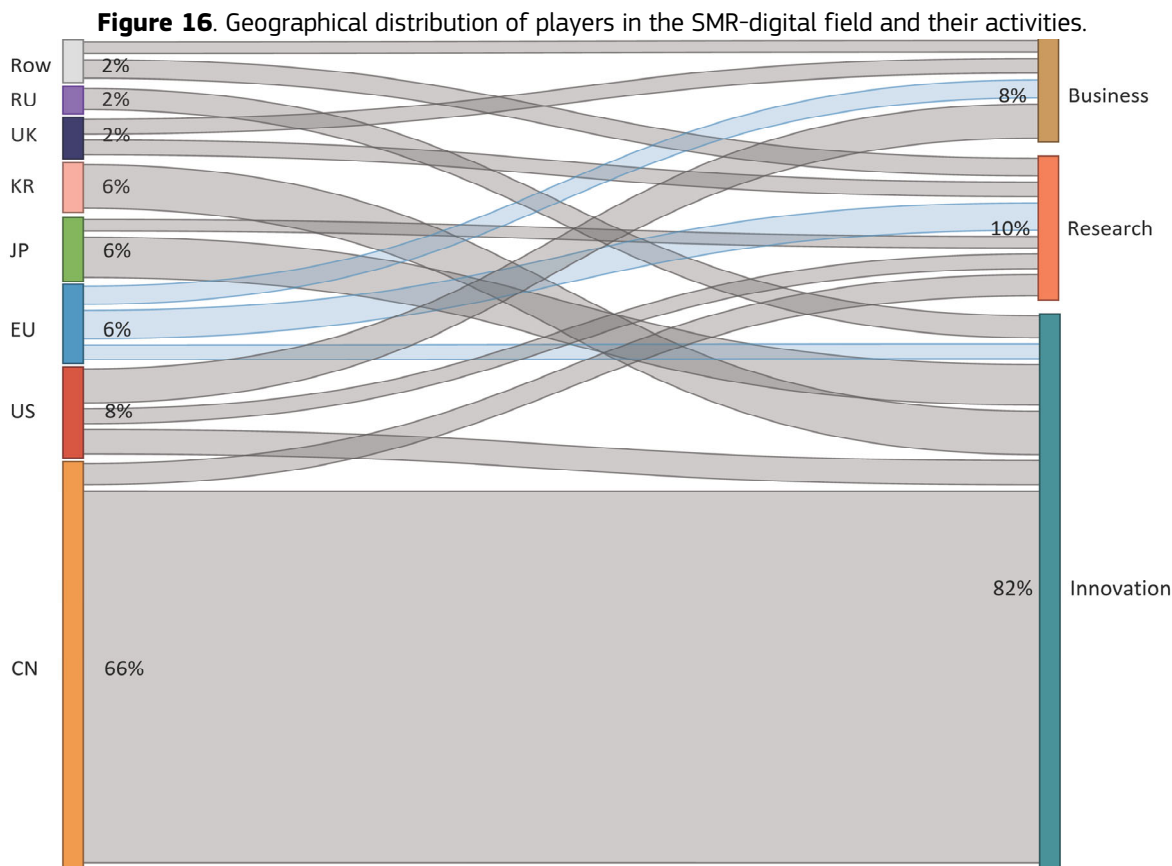
3.2 Increasing interest in SMRs by digital leaders

To close the innovation gap, Europe needs energy, including for its digital transformation. The frenzy of Artificial Intelligence (AI) development partly explains this rise. Data centres underpinning AI models require additional power. The International Energy Agency expects that global data centres alone will need more than double the electricity they currently consume, with estimated consumption around 945 TWh by 2030 (IEA, 2025a). In Europe –the EU plus Norway, Switzerland and the UK–, power demand for data centres is expected to grow to approximately 35 GW by 2030, up from 10 GW in 2024 (Granskog et al., 2024). To meet this new IT load demand, more than \$250 to \$300 billion of investment will be needed in data centre infrastructure, excluding power generation capacity.

This rise in energy demand for data-driven tools is boosting private investments from high-tech industries into SMRs. On 14 October 2024, Google announced the first corporate agreement to deploy multiple SMRs –six to seven reactors, according to the announcement– in the US, the first to be deployed by 2030. Google will power its data centres with 500 MWe of the Kairos Power’s fluoride salt-cooled high-temperature reactor. Governments and other big tech companies such as

Microsoft and Amazon are aware of the growth in energy demand and are also exploring investment strategies in clean energy technologies (The Economist, 2024; Yahoo Finance, 2024)³⁷.

At the same time, SMR development can benefit from digital transformation in areas like AI applied to monitoring safety networks and enhance cybersecurity (IAEA, 2023). The appearance of startups is also accelerating the digitalization of the nuclear industry. These startups use digital tools and technologies in their SMR designs, enabling a more efficient and streamlined development process (Tanarro et al., 2024).



Note: Geographical areas refer to China (CN), United States (US), European Union (EU), Japan (JP), South Korea (KR), United Kingdom (UK), Russia (RU) and rest of the world (RoW). Source: JRC TES database.

On the TES ecosystem, which captures SMR players up to 2023, most actors in the intersection with digital conduct innovation activities (**Figure 16**). China also leads in the SMR-digital intersection, where the Nuclear Power Institute of China and the Shanghai University stand out. Similar to the EIs overlap, the EU is the only major economy where the highest share of activities in the overlap belong to research. Politecnico di Milano or the Luxembourg Institute of Science and Technology are among the research centres at the crossroads of digital and SMRs.

³⁷ For a detailed summary of countries' support to SMR development, visit the World Nuclear Association's website: <https://world-nuclear.org/information-library/nuclear-fuel-cycle/nuclear-power-reactors/small-nuclear-power-reactors>

4 Conclusions

The global energy landscape is undergoing a significant transformation, with increasing demand for clean energy sources that at the same time mitigate vulnerabilities and enhance strategic autonomy. Small modular reactors (SMRs) have emerged as a promising solution, offering a flexible affordable alternative to traditional large nuclear reactors. As the EU strives to achieve its strategic economy objectives, SMRs have the potential to contribute firmly to the clean energy transition.

This report provides some details on the current situation in the SMR field and analyses the results of a first test run using the Techno-Economic ecoSystem (TES) methodology. A set of keywords identifying SMR reactor types, technologies and components lays the basis for the mapping of related activities across various databases. From the extracted data, TES builds a graph database that allows generating a network representation of the ecosystem.

Our TES analysis of the global and EU SMR ecosystem reveals a dynamic landscape, with a growing number of players, largely consisting of private firms. In total, there are approximately 4 000 players, when also accounting for research institutes, and government entities. China accounts for nearly 65% of the players and is particularly dominant in innovation. Overall, countries focus on patenting activities, highlighting the importance of registering designs and technologies prior to commercialization. While the EU boasts a robust research network and a strong presence in scientific publications, challenges remain in transforming research excellence into innovation and expanding the business network. This effect can be observed not only in nuclear but in many other technology innovations, where Europe is strong in innovation and regulation, but comparatively weak and slow in economic implementation.

The scoping study presented here goes beyond the JRC's classical assessments of nuclear safety and waste for SMR. It clearly shows the high potential and added value of the TES methodology to analyse the SMR ecosystem, specifically those techno-economic aspects that are crucial for assessing the credibility of certain SMR concepts and the involved supply chain.

Future application of TES to SMR will refine, broaden and update the exploration of activities and players, considering a selection of categories, and examine firm characteristics with the aim to ultimately suggest and support well-informed policy actions.

However, the development and deployment of SMRs also faces significant challenges. Amongst others, these include:

1. **Safety assessment capability and regulatory harmonisation:** Several projects at the IAEA (IAEA, 2023), the OECD-NEA (NEA, 2021), the Generation IV International Forum (its Risk and Safety Working Group³⁸) and several Euratom indirect actions (e.g., HARMONISE³⁹) are dealing with this subject. Harmonised regulatory frameworks at the international and EU level and clear licensing processes are required to enable the development and deployment of SMRs (NEA, 2022). In this area, the European Commission recommends an approach that was successfully tested on the early Nuward SMR design, namely cooperation between regulators of different EU Member States interested in deploying a specific SMR design.

³⁸ <https://www.gen-4.org/gif-activities/working-groups/risk-and-safety-working-group>

³⁹ <https://cordis.europa.eu/project/id/101061643>

2. **Uncertainty regarding cost and financing:** Nuclear energy requires significant upfront investment (IEA 2025b, World Nuclear Association⁴⁰, among others).

Several SMR demonstration projects are affected by large cost increases and uncertainties. These uncertainties and the related risks can only be removed once SMRs are built in sufficient numbers to ascertain real-world cost figures. Although the economic model of SMR is based on the economy of replication in place of the economy of size, thus reducing overnight cost, access to financing can still be a market entry barrier for developers. A number of existing public and private financing tools are either not open or unsuitable for nuclear demonstration and deployment projects. This is gradually changing with several EU Member States revising their nuclear strategy and even the European Investment Bank⁴¹ opening to nuclear projects.

3. **Public engagement:** The attitude of the public regarding nuclear energy is varying strongly across EU Member States. The debate is mostly on safety and waste, both subjects are strong elements of the Euratom direct actions performed by the JRC. The scientific technical results can support a public discussion based on facts.
4. **Technology readiness level and timelines:** The large number of SMR concepts are at very different levels of technology readiness. Several of them claim safety and performance features that still require analytical confirmation and ultimately demonstration. still a relatively recent technology, and there are risks associated with their development, deployment, and operation, including the potential for accidents and cybersecurity threats. Some of their claimed features still need to be proved in practice. There is also room to improve the risk of time and cost overruns.

⁴⁰ <https://world-nuclear.org/information-library/economic-aspects/financing-nuclear-energy>

⁴¹ <https://www.euronews.com/business/2024/02/27/eib-president-backs-nuclear-power-as-she-outlines-new-vision>

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List of abbreviations

Abbreviation	Definition
AI	Artificial intelligence
AMR	Advanced Modular Reactor
ARIS	Advanced Reactor Information System
BWR	Boiling water reactor
CH	Switzerland
CN	China
CO ₂	Carbon dioxide
EC	European Commission
EeEU	Europe excluding the EU
EElS	Energy-intensive industries
EU	European Union
Euratom	European Atomic Energy Community
FNR	Fast Neutron Reactor
FP7	Seventh Framework Programme
GDP	Gross domestic product
Gen-IV	Fourth generation nuclear systems
GFR	Gas-cooled fast reactor
GIF	Generation IV International Forum
GW	Gigawatt
GWe	Gigawatt electric
H2020	Horizon 2020
HTGR	High-temperature gas-cooled reactor

Abbreviation	Definition
HTR	High-temperature reactor
IAEA	International Atomic Energy Agency
IEA	International Energy Agency
IPHWR-220	Indian Pressurized Heavy Water Reactor-220
IT	Information technology
JP	Japan
KR	South Korea
kWh	kilowatt-hours
JRC	Joint Research Centre
LFR	Lead-cooled Fast Reactor
LMR	Liquid metal cooled reactors
LTO	Long-term operation
LWR	Light water reactor
MIT	Massachusetts Institute of Technology
MMR	Micro modular reactor
MS	EU Member States
MSR	Molten salt reactor
Mt	Million tonnes
MWe	Megawatt electric
NEA	Nuclear Energy Agency
NECP	National Energy and Climate Plan
NZIA	Net Zero Industry Act

Abbreviation	Definition
OECD	Organisation for Economic Co-operation and Development
OECD-NEA	OECD-Nuclear Energy Agency
PHWR	Pressurised heavy-water reactor
PWR	Pressurized water reactor
R&D	Research and development
R&I	Research and innovation
RoW	Rest of the world
RU	Russia
SCWR	Super critical water-cooled reactor
SET Plan	Strategic Energy Technology Plan
SFR	Sodium-cooled Fast Reactor
SMR	Small Modular Reactor
SNETP	Sustainable Nuclear Energy Technology Platform
TES	Techno-Economic ecoSystem
TRL	Technology readiness level
TWh	Terawatt hour
UK	United Kingdom
UN	United Nations
US	United States
VHTR	Very high temperature reactor
WNA	World Nuclear Association

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Annex 1

Table 2. Keyword list used to map the SMR ecosystem.

Additional Heat	GE Hitachi	LOFA	Power Output
Alternative coolant	BWRX-300	Loss-of-heat sink	Pressure vessel
Blue Capsule	Gen IV SMR	LOHS	PHWR
BWR	Generation IV SMR	Low pressure conduction cooldown	Pressurised heavy water reactor
Boiling water reactor	GWe	LEU	PWR
Brownfield site	Gigawatt electric	Low-enriched uranium	Pressurized water reactor
CADOR core	Graphite-moderated high temperature reactor	LWR technology	Radiological risk
Calogena	Heat removal	MWe	Reactivity insertion accident
CNSC	Hexana	Megawatt electric	RIA
Canadian Nuclear Safety Commission	HALEU	Metal-cooled system	Refuelling interval
Capex	High-assay low-enriched uranium	MMR	Remote control
Closed fuel cycle	HTGR	Micro modular reactor	Research Centre Rez
CNP-300	High Temperature Gas- cooled Reactor	MOX	RITM-200M
Co-electrolysis	High-temperature gas- cooled SMR	Mixed oxide fuel	Rolls-Royce
Conventional fuel	HTR	Mobile SMR	Safety system
Coolant	High-temperature reactor	Transportable SMR	Seaborg
Copenhagen Atomics	HTGR-POLA	Modular construction	SEALER
Core heat	Hybrid energy system	Modularity of fabrication	Semi-autonomous operation
CVR	HES	Module factory fabrication	Single-unit LWR-SMR
Cycle length	Hydrocarbon chain	MSR	Small light-water reactor
Demineralization	Hydromine	Molten salt reactor	Small LWR
DW	Improved transportability	Multi-module LWR-SMR	SMR
Demineralized water	IAEA	Multi-module Pressurized Water Reactor	Small modular reactor
Desalination	International Atomic Energy Agency	Naarea	Small reactor
Distributed generation	IEA	NEA	SNP350

District heating	International Energy Agency	Nuclear Energy Agency	SFR
Dual Fluid Reactor	JIMMY	Newcleo	Sodium-cooled Fast Reactor
EcoSMR	KLT-40S	Non-LWR technology	Na-HTR
eFuel	LDR-50	Nuclear fuel	Sodium-cooled High Temperature Reactor
eKerozene	LFR	NLC	Stand-alone unit
Electrolysis facility	Lead-cooled Fast Reactor	Nuclear Law Committee	Steam cycle
Emergency cooling	DFR	Nuclear power development	Steam generator
EPZ	Lead-cooled Molten Salt Reactor	Nuclear power plant design	Stellaria
Emergency planning zone	LCOE	Nuclear reactor technology	Structural material
eMethane	Levelised cost of electricity	NRC	Synfuel
eMolecule	Levelised cost of heat	Nuclear Regulatory Commission	TRL
Energy-dense vector	LCOH	Nuclear research and development	Technology readiness level
Enriched uranium	Levelised cost of hydrogen	Nuclear safety	Teplator
Fast molten salt reactor	LCOH2	NSSS	Thermal molten salt reactor
Fast neutron reactor	LRL	Nuclear steam supply system	Thermal neutron
Fast-neutron SMR	Licensing readiness level	Nuclear technology	Thermal Output
FNPP	LWR	Nuclear waste	Thermal power
Floating nuclear power plant	Light water reactor	NuScale Power Module	THORIZON
Floating reactor	Light water SMR	NuScale Power	USNC
Forced circulation	Liquid metal	NUWARD	VRE
Fuel Enrichment	Liquid metal fast reactor	Off-grid operation	Variable renewable energy
Fuel form	LOCA	Opex	VOYGR-12
Fuel processing	Loss-of-coolant accident	Passive safety	X-energy Xe-100
Gas-cooled system	Loss-of-flow accident	Plutonium stockpile	Zero-carbon transition

Annex 2

Table 3. Top-3 or largest players by number of activities by EU country.

Country/Player	Activities
France	714
COMMISSARIAT A L ENERGIE ATOMIQUE ET AUX ENERGIES ALTERNATIVES	98
INSTITUT DE RADIOPROTECTION ET DE SURETE NUCLEAIRE	79
ELECTRICITE DE FRANCE	78
Germany	673
KARLSRUHER INSTITUT FUER TECHNOLOGIE	94
GESELLSCHAFT FUR ANLAGEN UND REAKTORSICHERHEIT (GRS) GmbH	53
HELMHOLTZ-ZENTRUM DRESDEN-ROSSENDORF EV	53
Italy	465
AGENZIA NAZIONALE PER LE NUOVE TECNOLOGIE, L'ENERGIA E LO SVILUPPO ECONOMICO SOSTENIBILE	89
POLITECNICO DI MILANO	68
ANSALDO NUCLEARE SPA	29
Belgium	360
JRC -JOINT RESEARCH CENTRE- EUROPEAN COMMISSION	107
STUDIECENTRUM VOOR KERNENERGIE / CENTRE D'ETUDE DE L'ENERGIE NUCLEAIRE	77
TRACTEBEL ENGINEERING	33
Spain	305
CENTRO DE INVESTIGACIONES ENERGETICAS MEDIOAMBIENTALES Y TECNOLOGICAS	78
UNIVERSIDAD POLITECNICA DE MADRID	30
EMPRESARIOS AGRUPADOS INTERNACIONAL SA	21
The Netherlands	257
NUCLEAR RESEARCH AND CONSULTANCY GROUP	71
TECHNISCHE UNIVERSITEIT DELFT	26
TECHNISCHE UNIVERSITEIT EINDHOVEN	11
Czechia	198
UJV REZ AS	68
CENTRUM VYZKUMU REZ SRO	44
STATNI USTAV RADIACNI OCHRANY v.v.i.	13
Sweden	190
KUNGLIGA TEKNISKA HOEGSKOLAN	64

Westinghouse Electric Sweden AB	34
CHALMERS TEKNISKA HOGSKOLA AB	28
Finland	133
TEKNOLOGIAN TUTKIMUSKESKUS VTT OY	74
FORTUM POWER AND HEAT OY	14
LAPPEENRANNAN-LAHDEN TEKNILLINEN YLIOPISTO LUT	12
Romania	111
REGIA AUTONOMA TEHNOLOGII PENTRU ENERGIA NUCLEARA - RATEN	41
RAAN DROBETA TURNU SEVERIN	17
UNIVERSITATEA POLITEHNICA DIN BUCURESTI	7
Poland	92
NARODOWE CENTRUM BADAN JADROWYCH	24
Warsaw University of Life Sciences (SGGW)	7
INSTYTUT CHEMII I TECHNIKI JADROWEJ	6
Hungary	92
ENERGIATUDOMANYI KUTATOKOZPONT	31
BUDAPESTI MUSZAKI ES GAZDASAGTUDOMANYI EGYETEM	18
MAGYAR TUDOMANYOS AKADEMIA KFKI ATOMENERGIA KUTATOINTEZET	15
Slovakia	62
VUJE AS	21
SLOVENSKA TECHNICKA UNIVERZITA V BRATISLAVE	15
INZINIERSKA VYPOCTOVA SPOLOCNOST TRNAVA S.R.O.	10
Lithuania	57
LIETUVOS ENERGETIKOS INSTITUTAS	41
KAUNO TECHNOLOGIJOS UNIVERSITETAS	8
VALSTYBINIS MOKSLINIU TYRIMU INSTITUTAS FIZINIU IR TECHNOLOGIJOS MOKSLU CENTRAS	4
Slovenia	54
INSTITUT JOZEF STEFAN	32
KEMIJSKI INSTITUT	7
Elektroinstitut Milan Vidmar	6
Denmark	50
DANMARKS TEKNISKE UNIVERSITET	9
Technical University of Denmark	6
Seaborg Technologies	5

Bulgaria	47
INSTITUTE OF NUCLEAR RESEARCH AND NUCLEAR ENERGY - BULGARIAN ACADEMY OF SCIENCES	22
Technical University of Sofia	14
Austria	36
TECHNISCHE UNIVERSITAET WIEN	5
MICROINNOVA ENGINEERING GMBH	3
UNIVERSITAET FUER BODENKULTUR WIEN	3
Portugal	27
INSTITUTO TECNOLÓGICO E NUCLEAR	4
Faculdade de Engenharia do Porto	3
Greece	21
ETHNIKO KENTRO EREVNAS KAI TECHNOLOGIKIS ANAPTYXIS	5
ARISTOTELIO PANEPISTIMIO THESSALONIKIS	3
Ireland	10
OxyMem	2
CELIGNIS LIMITED	2
Estonia	6
TARTU ULIKOOL	3
Croatia	4
University of Rijeka	1
SVEUCILISTE U ZAGREBU, FAKULTET STROJARSTVA I BRODOGRADNJE	1
Inetec	1
Luxembourg	2
Luxembourg Institute of Science and Technology	1
ARISTENG SARL	1

Source: JRC TES database.

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